Lighthouse

AI agent orchestration & decision intelligence

DEMO & USER GUIDE

Four agents. Four owners. One decision.

How to open, use and present Lighthouse: the governance layer that reconciles a Sales, Finance, Operations and Legal AI agent for a 50-store Walmart U.S. pilot. Each agent is owned by a department lead, and a named executive makes the final call.

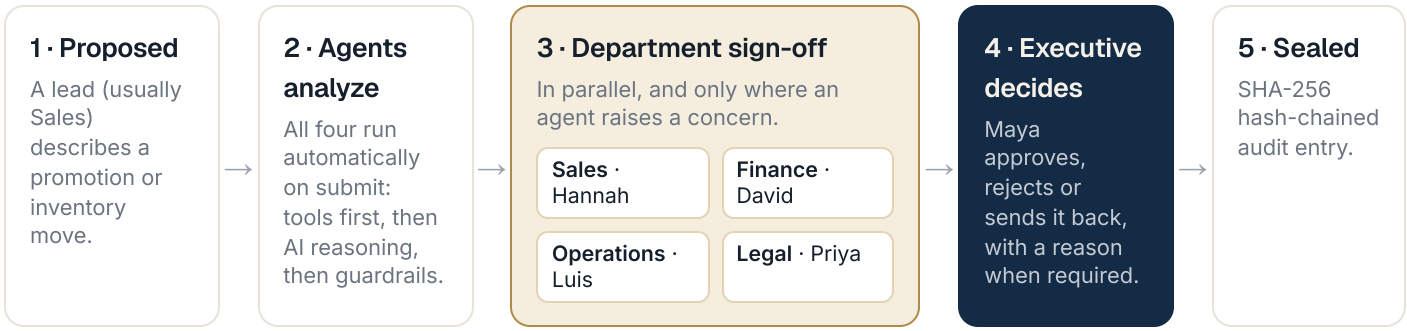
Live app	`(add after deployment)`
Code	`github.com/BhavyaAk25/lighthouse`
Sign in	Pick a person. No passwords. The administrator uses PIN `4210`.
Cost	$0: Vercel Hobby, Neon free tier, Gemini and Groq free tiers.

ISTM 6210 / 4210 Capstone · Fall 2026 · The George Washington University
Ajay · Bhavya · Ehson · Jaehoon · Kunal · Wilfred · Faculty advisor: Dr. Soheil Sibdari
Academic prototype, not affiliated with Walmart Inc. Demand data is public Kaggle Walmart data; everything generated is labelled.

1 · THE IDEA

Lighthouse in one minute

Companies now run several specialised AI agents. Each can be right on its own and still disagree with the others: Sales sees demand, Finance sees a loss, Operations sees a full distribution center, Legal sees a risky claim. **Lighthouse is the layer above the agents.** Every agent answers in the same format with cited evidence. The department that owns the agent signs off on it, Lighthouse reconciles the four positions, and the executive decides. Every step lands in a tamper-evident log.



2 · THE PEOPLE

Who does what

PERSON	ROLE	DOES	CANNOT
MC **Maya Chen**	Executive · decision owner	Sees the full packet and all four sign-offs. Approves, rejects or sends back. Can overrule a Sales, Finance or Ops block with a reason, and can decide when a sign-off is overdue.	Overrule Legal; approve before required sign-offs
HB **Hannah Brooks**	Sales lead	Proposes promotions (submitting runs all four agents). Owns the Sales agent: demand, pricing, channel.	Sign for other departments
DO **David Okafor**	Finance lead	Owns the Finance agent: margin, ROI, break-even. Confirms, adjusts or blocks its answer.	Decide the case
LM **Luis Moreno**	Operations lead	Owns the Operations agent: inventory, DC capacity, fulfillment. Also proposes inventory moves.	Decide the case
PR **Priya Raman**	Legal lead	Owns the Legal agent: claims, privacy, state rules. Approves, approves with conditions, or rejects. **Her rejection is final.**	Approve a case a rule prohibits
JE **Jordan Ellis** `PIN 4210`	System administrator	Edits rules, thresholds and the Live AI / Replay setting (each change needs a reason). Resets the demo, tests providers, runs the test lab.	Sign off or decide (separation of duties)

3 · THE RULES

Why it is built this way, and the rules

Ownership

Each department lead owns their agent. They can re-run it, confirm it, change its answer with a reason, or block it. Nobody runs another department's agent.

Management by exception

A lead only signs when their agent says hold or reject, needs a review, is under 65% confident, or lacks evidence. In the 12 demo cases, about two out of three lanes clear automatically.

Accountability

The packet shows what the AI said and what each person signed. The executive can overrule Sales, Finance or Operations with a written reason. Legal can never be overruled.

A lead must sign when their agent…

- says Hold or Reject
- says Review required (for example, Legal flags a claim)
- is less than **65%** confident (admin can change this)
- is missing evidence (the output is marked invalid, never imputed)

Otherwise the lane shows *Cleared automatically*. A lead can still step in on any lane.

What a lead can do

- **Confirm** the agent's answer (a note is optional)
- **Adjust** it to proceed / with conditions / hold: needs a 20+ character reason, and lowers Lighthouse's confidence for that agent by 15%
- **Block** it: needs a reason. Legal's version is *Reject*.
- **Re-run my agent** at any time. Its old sign-off is cleared and must be given again.

When the executive can approve

- Every lane is cleared or signed, **and**
- Legal has not rejected it and no legal rule prohibits it
- A Sales, Finance or Ops **block** can be overruled with a written reason, and is recorded as an override
- Going against Lighthouse's recommendation, or a high-risk case, also needs a reason

Reject and Send back are allowed any time after the agents run.

Deadlines and escalation

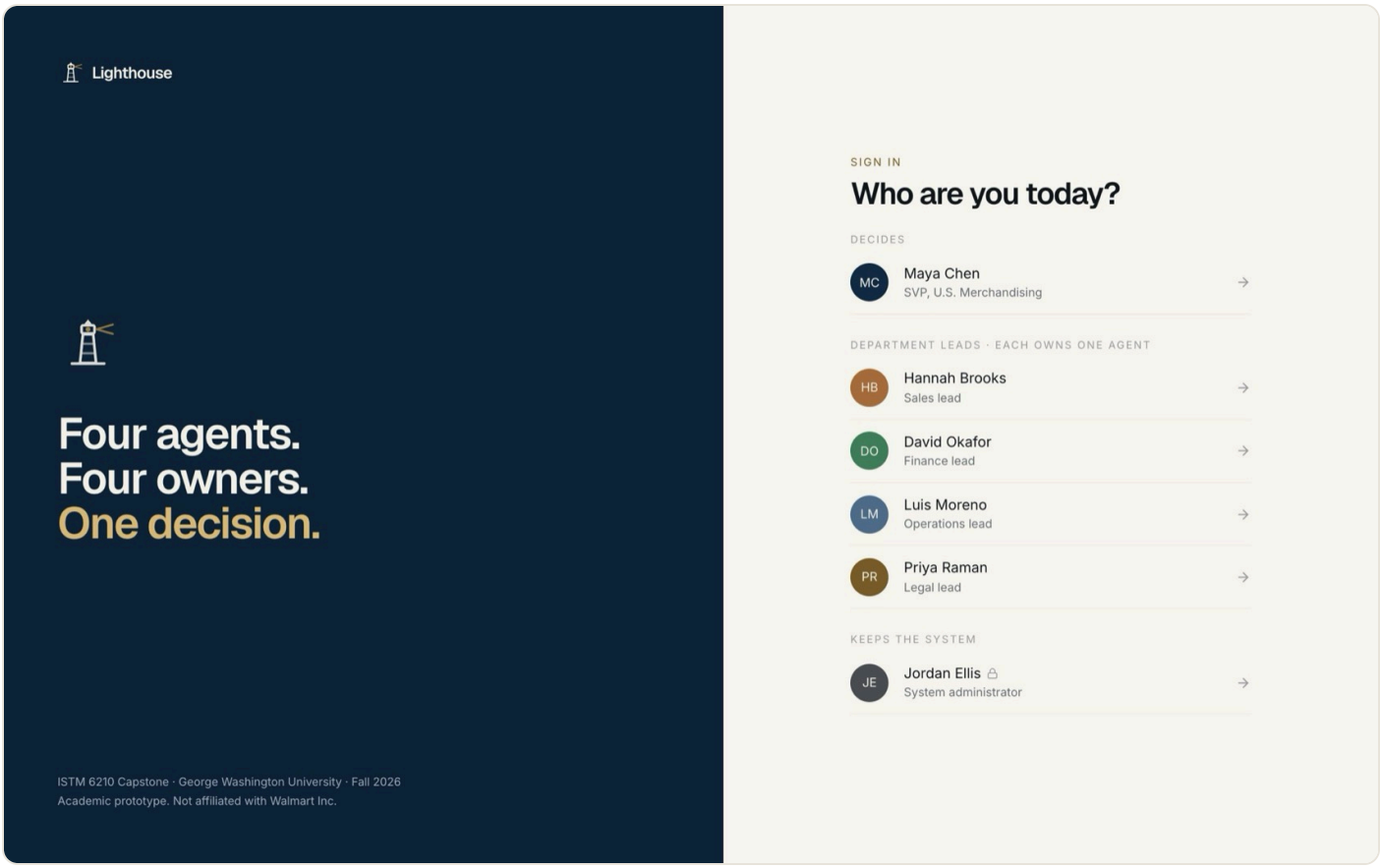
- Each sign-off is due **48 hours** after the agents run (admin can change this)
- An overdue Sales, Finance or Ops lane escalates: the executive may decide without it, with a reason, and the record names who didn't sign
- An overdue **Legal** lane never escalates. Approval stays locked.

Lighthouse reconciles twice: first on the four AI answers, then again on what the departments actually signed. When the signed result differs, the packet says so in plain words: *"The AI agents alone said hold (75%). Department sign-offs changed it."*

4 · GETTING READY

Before class (10 minutes)

- [ ] Open the live app. Sign in as **Jordan Ellis** (PIN 4210).
- [ ] *Rules & agents → Demo controls → Reset demo cases*. Do this **on demo day**: sign-off deadlines run on real time, so a reset done days earlier makes every lane overdue.
- [ ] *Rules & agents → Agents & AI → Test providers*. Both Gemini and Groq should show OK.
- [ ] Set the top-right switch: **Live AI** if the network is good, **Replay** if it isn't (saved AI answers, identical results). The setting applies to everyone in that browser.
- [ ] Tip: give each teammate one persona, so the hand-offs feel like real people. Switching person is one click (*Switch*, bottom left).



Sign-in groups people by what they do: the executive, the four department leads, and the administrator.

5 · THE DEMO

Demo script (about 8 minutes)

The story follows one promotion from proposal to sealed decision, with a detour that shows a human disagreeing with their AI.

TIME	PERSON	WHAT THE AUDIENCE SEES
0:00–0:45	Anyone	The problem, the five roles (sign-in screen)
0:45–2:15	Hannah · Sales	Submits D-2026-001; four agents run live; she lands on her own lane, cleared automatically
2:15–3:15	Priya · Legal	Only Legal is needed; she approves the claim with conditions
3:15–4:30	David · Finance	D-2026-003: his agent says reject; he re-runs it, then changes it to "proceed with conditions" with a reason
4:30–6:15	Maya · Executive	Her desk; D-2026-003 shows "AI said hold, sign-offs changed it"; D-2026-006 escalated; approves D-2026-001
6:15–7:15	Maya	Flow (four parallel lanes) and the Dashboard
7:15–8:00	Jordan · Admin	Audit chain verification and the 18 acceptance tests

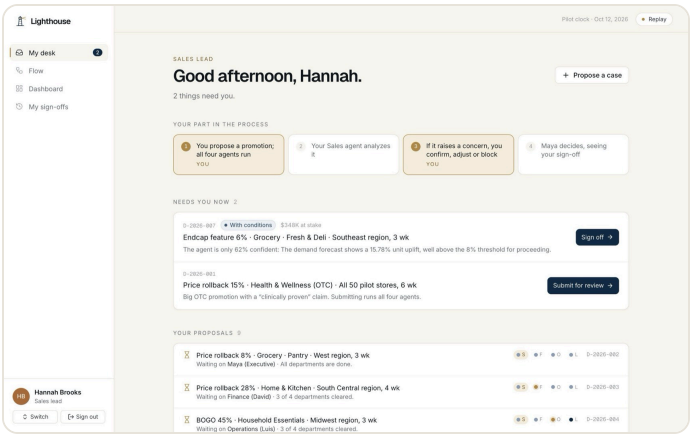
1 **Set the scene.** SIGN-IN SCREEN

> *"A promotion decision touches four departments, and each now has an AI agent. Lighthouse gives each department ownership of its agent, asks a person to sign only when their agent has a concern, and leaves the final call to a named executive."*

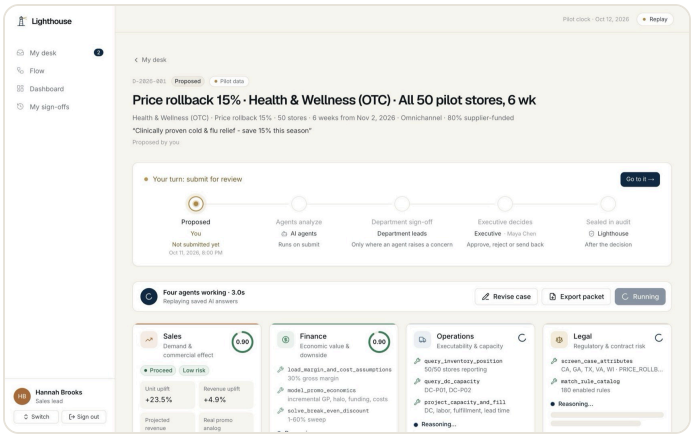
2 **Hannah proposes and submits.** HANNAH · SALES LEAD

Her desk shows two things: a sign-off on D-2026-007 and her draft D-2026-001 (15% OTC rollback, all 50 stores, claim "Clinically proven cold & flu relief"). Click **Submit for review**, then **Submit for review** again on the case.

> *"Submitting runs all four agents in parallel. Each one first calls tools against the database (you can see the tool names), then the model reasons over that evidence, then guardrails check it."*

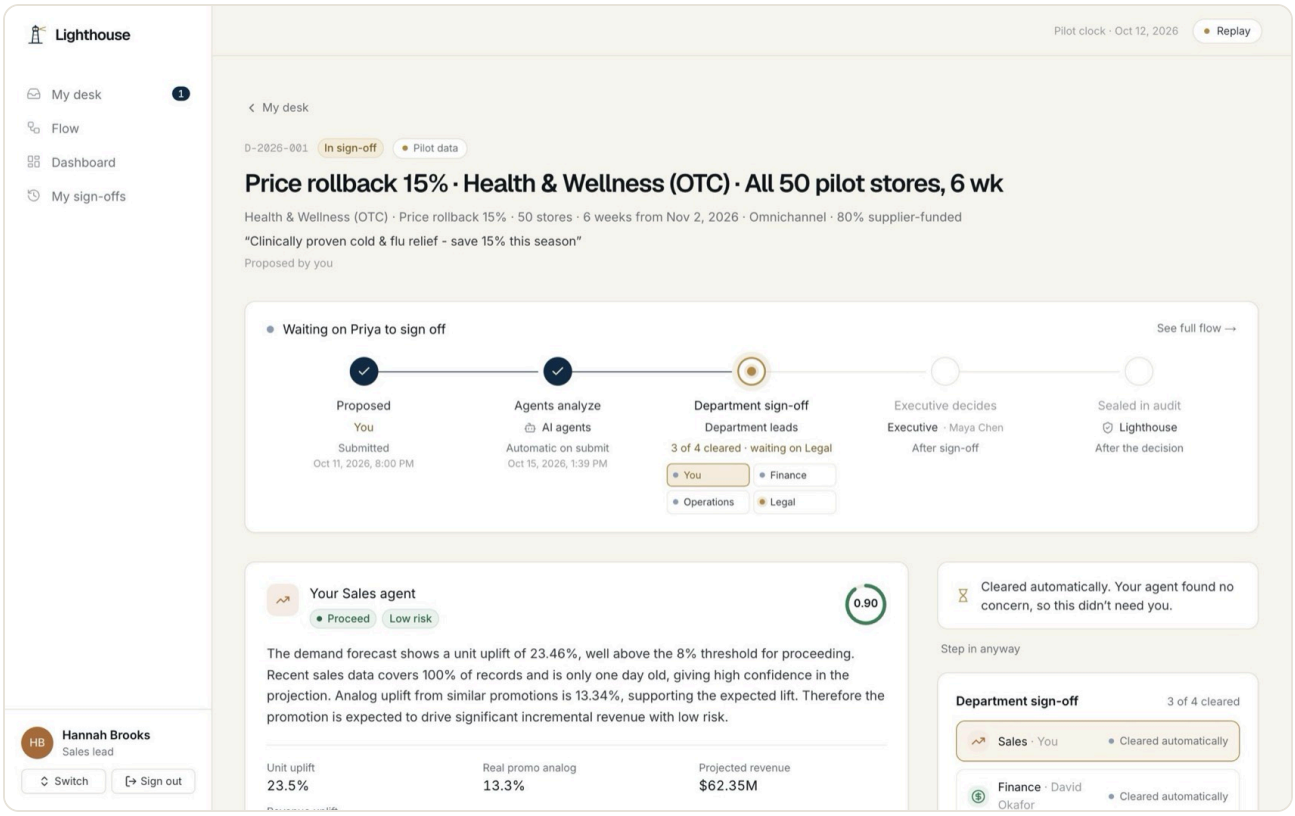


Hannah's desk: only her work, each with the reason it needs her.



After Submit: four agents mid-run, tool calls first, then reasoning.

When the run finishes, Hannah lands on **her own lane**: the Sales agent says proceed (+23.5% unit uplift), so it *cleared automatically*. The journey bar shows *3 of 4 cleared · waiting on Legal*.

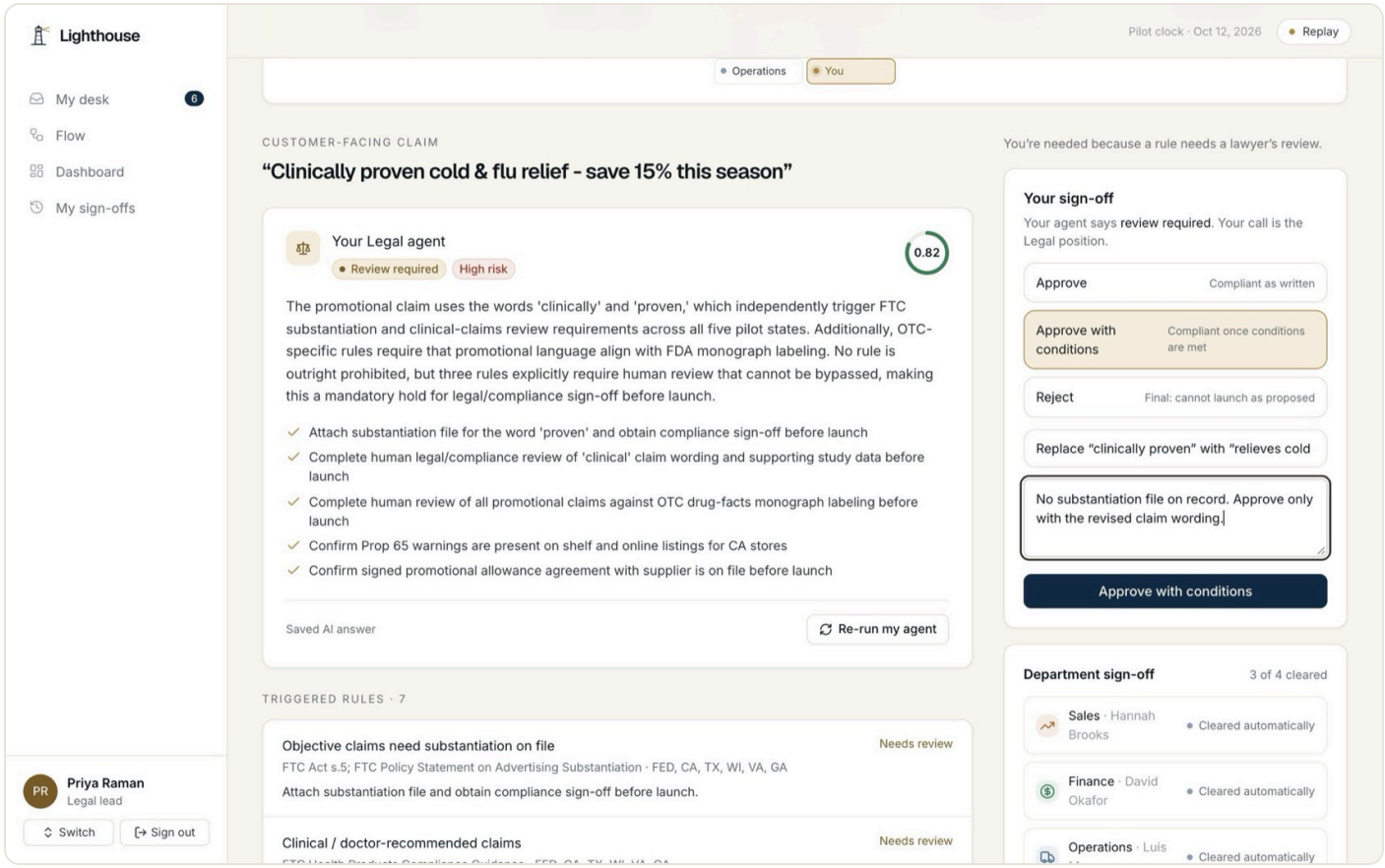


After submitting: the journey bar shows four parallel lanes. Sales, Finance and Operations cleared automatically; Legal is waiting.

3 **Priya signs off on the claim.** PRIYA · LEGAL LEAD

Her desk lists the cases where the Legal agent flagged a rule. Open D-2026-001: the claim, her agent's finding and 7 triggered rules (FTC substantiation, OTC monograph). Choose **Approve with conditions**, add the condition and a note, and sign.

> *"Legal is the only department needed here. Priya doesn't review the other three, and they don't wait on her. Her rejection would be final; even the executive can't overrule it."*

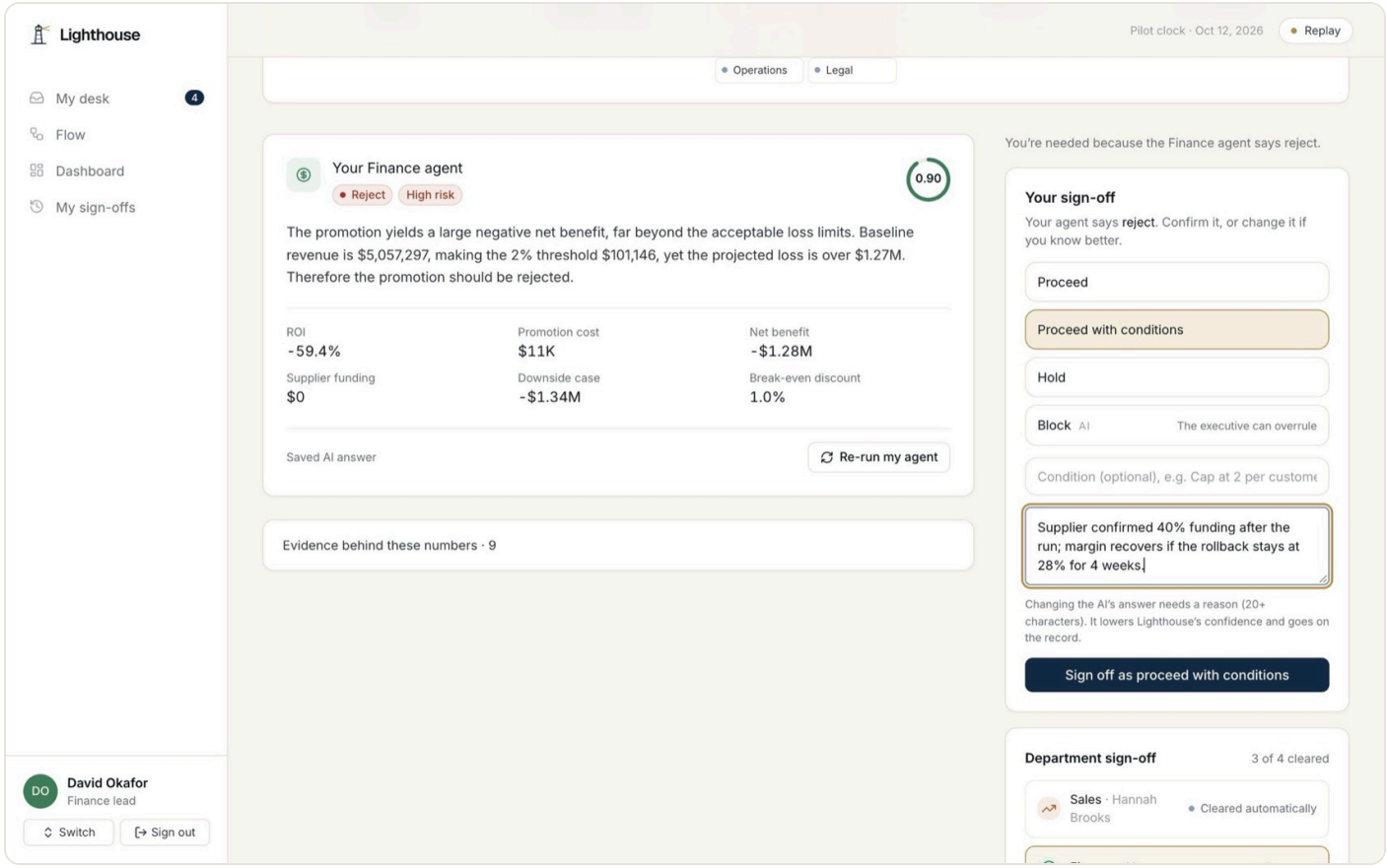


Priya's lane: her agent's finding, the triggered rules, her sign-off form, and the status of the other three departments.

4 **David disagrees with his agent.** DAVID · FINANCE LEAD

Open D-2026-003 (28% Home & Kitchen rollback). His agent says Reject: net benefit −$1.28M. Click **Re-run my agent** to show that he owns it (it re-runs in a few seconds; the others keep their answers). Then choose **Proceed with conditions**, type the reason ("Supplier confirmed 40% funding after the run…") and sign.

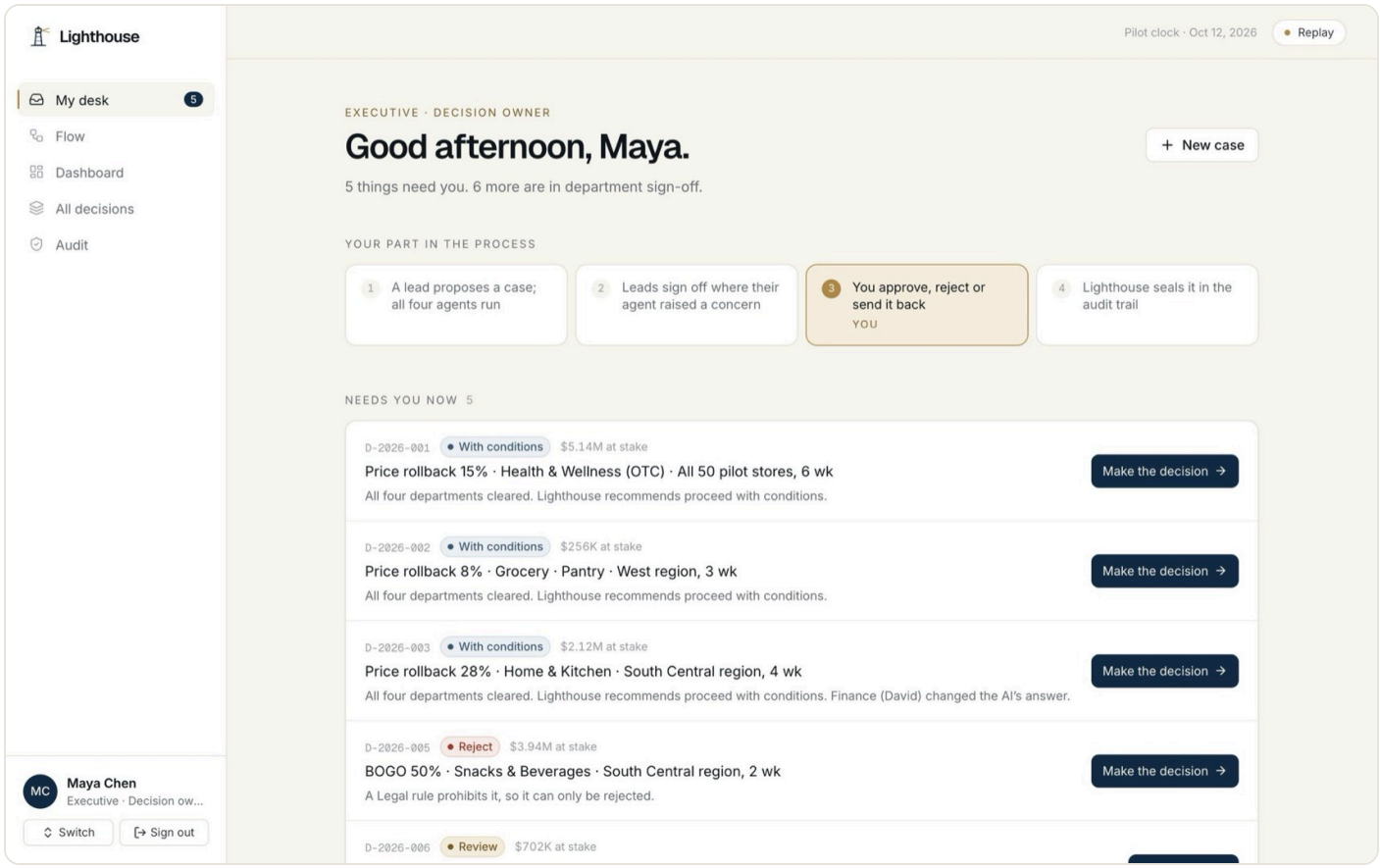
> *"The AI is advice. The department head is accountable. If David changes his agent's answer he has to say why, it costs Lighthouse confidence, and both versions stay on the record."*



David's sign-off: the AI's answer is marked "AI". Changing it requires a reason.

5 **Maya's desk.** MAYA · EXECUTIVE

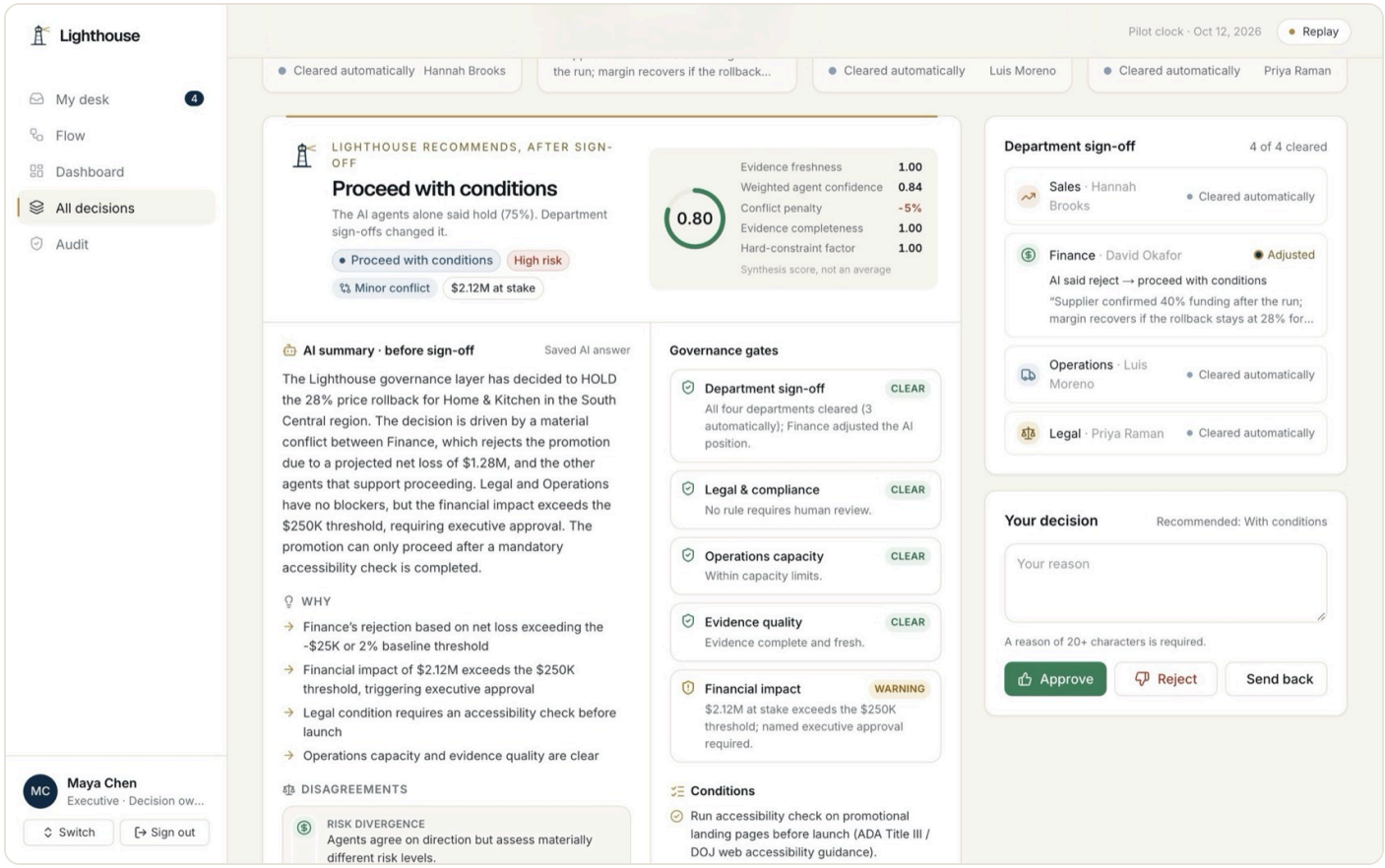
Five things need her: cases where every department has cleared, the prohibited Texas seltzer BOGO (D-2026-005, which can only be rejected), and D-2026-006, where Operations missed the deadline. Below that she sees what is still in sign-off and with whom.



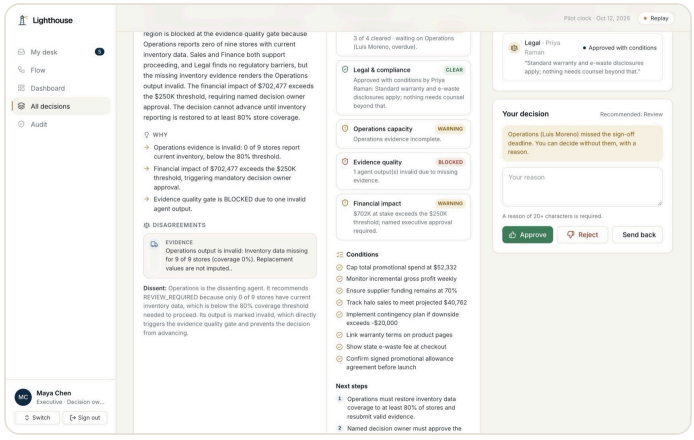
The executive's desk: decisions ready for her, each with a one-line reason.

6 **Human vs AI, side by side.** Open D-2026-003. Each agent card has a footer saying who owns it and what they did. The packet reads *“Lighthouse recommends, after sign-off: Proceed with conditions. The AI agents alone said hold (75%). Department sign-offs changed it.”* The sign-off panel shows *AI said reject → proceed with conditions* with David’s reason.

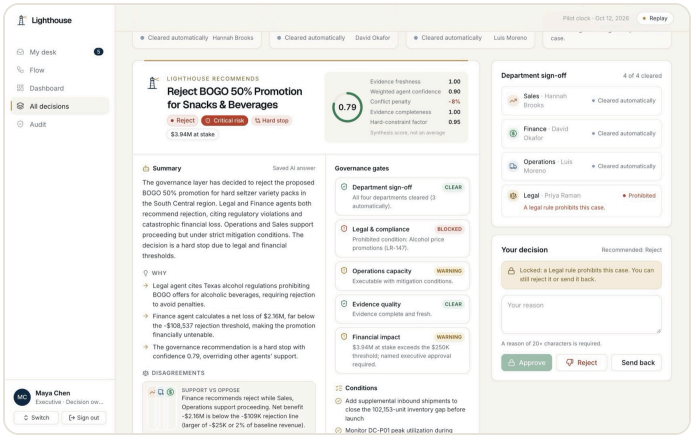
> *“Nothing is averaged away. Maya sees exactly where a person overruled their AI, and why, before she decides.”*



The reconciled packet after sign-off, the four department lanes, and the decision panel.

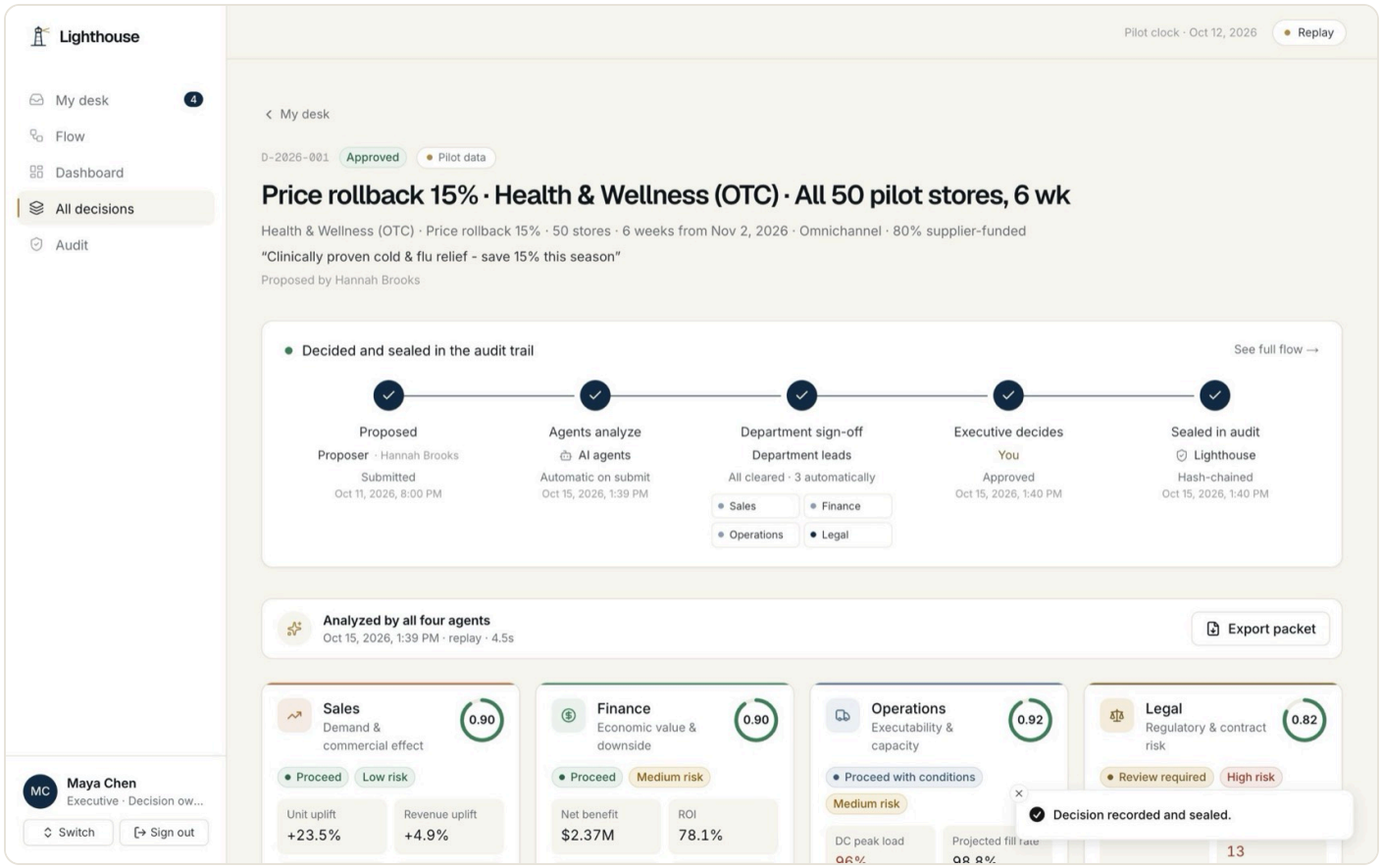


D-2026-006: Operations is overdue, so Maya may decide without it (with a reason). Legal would never escalate.



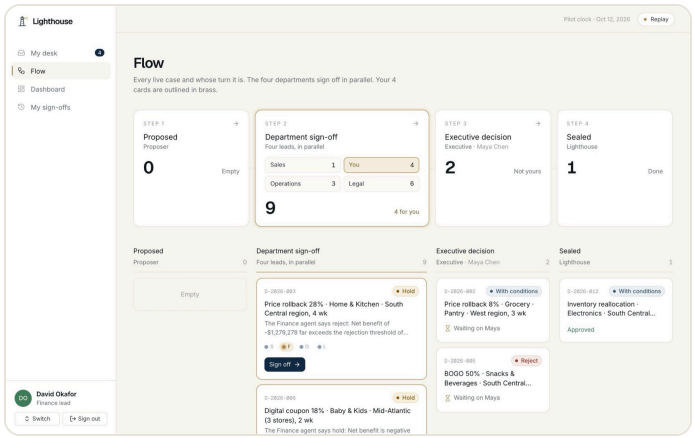
D-2026-005: a Texas rule prohibits the alcohol BOGO. Approval is locked for everyone.

7 **Approve D-2026-001.** Open it, type a reason ("Approved with Legal's revised claim; supplier funds 80%…") and click **Approve**. Try it first with an empty reason: it is refused, because the case is high-risk. The journey bar turns to *Decided and sealed in the audit trail*.

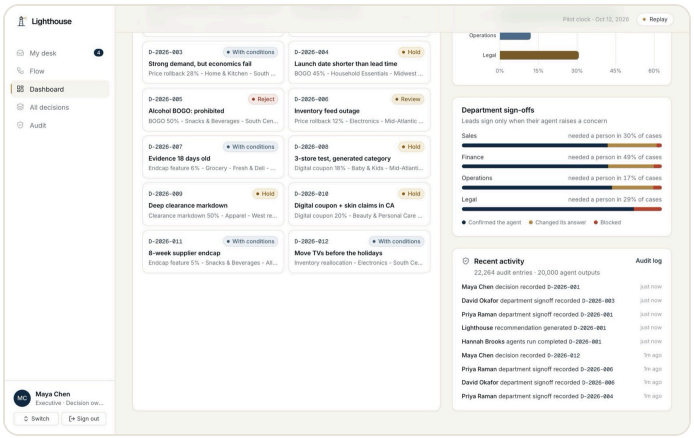


Every step done: proposed, analyzed, signed off (three automatically), decided, sealed.

8 **Flow and Dashboard.** *Flow* shows every live case by stage. The sign-off stage is split into the four departments with their open counts, and the signed-in person's cards glow brass. The *Dashboard* shows the pilot across 5,000 decisions, including how often each department needed a person and how often they agreed with their agent.



Flow: four stages, with department sign-off running in parallel.

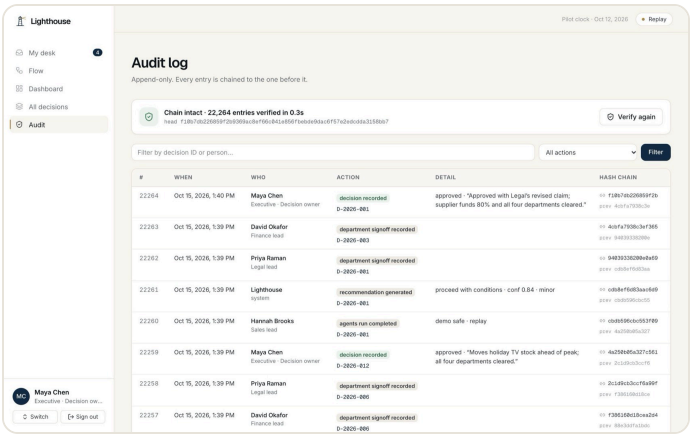


Dashboard: department sign-offs (confirmed, changed, blocked) and recent activity.

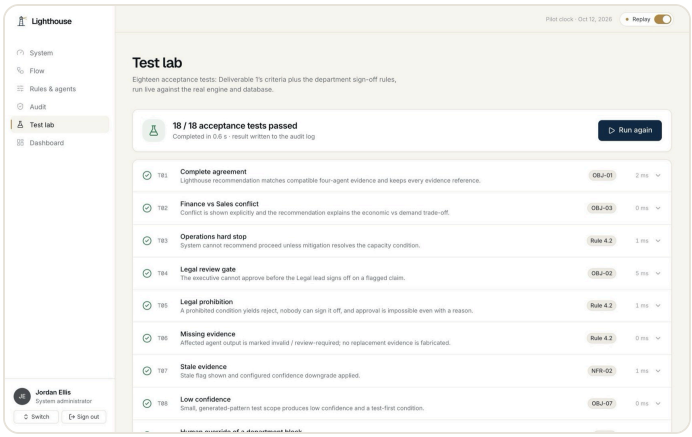
9 Audit and tests. JORDAN · ADMIN

Audit → Verify chain re-hashes about 22,000 entries in under a second. *Test lab → Run all tests* runs 18 acceptance tests live: Deliverable 1's criteria plus the sign-off rules (management by exception, escalation without skipping Legal, human adjustments costing confidence).

> *"If anyone edited a past decision or sign-off, verification would point to the exact entry that changed."*

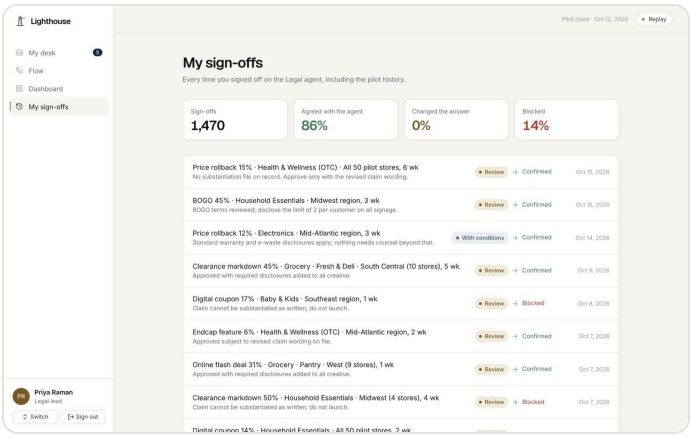


Tamper-evidence check over the full audit chain.

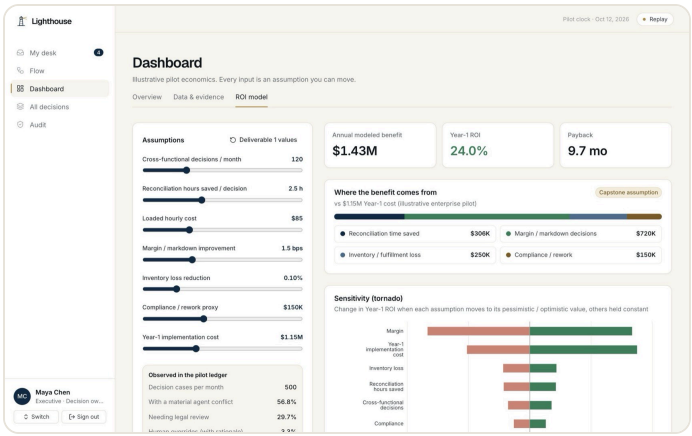


18 of 18 acceptance tests passing against the real engine and database.

> **If you have extra time:** as Priya, open *My sign-offs* (how often she agreed with her agent); as Jordan, change the sign-off confidence floor with a reason and show the new audit entry; open *Dashboard → Data* and *ROI*.



My sign-offs: each lead's record of confirming, changing or blocking their agent.



ROI model (Deliverable 1: $1.43M benefit, 24% ROI, 9.7-month payback) with live sliders.

6 · REFERENCE

The twelve demo cases after a reset

With Replay on, outcomes are exactly as listed. Live AI wording varies a little from run to run, but the governance rules fix the outcome.

CASE	SCENARIO	STARTING STATE	WHO MUST SIGN · WHAT IT SHOWS
D-2026-001	15% OTC rollback, “clinically proven” claim	Proposed (submit live)	Legal only. The anchor case: legal gate, then approval.
D-2026-002	8% pantry rollback, supplier-funded	With executive	Nobody: all four cleared automatically. With conditions
D-2026-003	28% Home & Kitchen rollback	In sign-off	Finance. Demand strong, economics fail: Hold. Best case for a human adjustment.
D-2026-004	BOGO launching in 3 days	In sign-off	Operations (Legal already signed). Lead time beats launch: hard stop, Hold.
D-2026-005	Hard seltzer BOGO in Texas	With executive	Nobody: a legal rule prohibits it. Reject, approval impossible.
D-2026-006	Electronics during an inventory feed outage	In sign-off, Ops overdue	Operations is 54h overdue, so it escalates to Maya. Evidence invalid, never imputed.
D-2026-007	Deli endcap on 18-day-old evidence	In sign-off	Sales, Operations, Legal: stale data drops every agent below 65% confidence.
D-2026-008	3-store baby digital coupon	In sign-off	Finance (hold) and Legal (children’s data).
D-2026-009	50% apparel clearance	In sign-off	Finance (reject) and Legal (reference-price rules).
D-2026-010	Beauty coupon, skin-repair claim, California	In sign-off	Finance and Legal (privacy plus cosmetic-drug claim).
D-2026-011	8-week supplier endcap, all stores	In sign-off	Legal (exclusivity terms).
D-2026-012	Move 6,000 TVs before the holidays	Approved	Proposed by Luis (Operations); shows a sealed case.

Desk counts after a reset: Maya 3 · Hannah 2 · David 4 · Luis 3 · Priya 5.

7 · RELIABILITY

Live AI, Replay and the backup plan

LAYER	WHAT IT IS	WHEN IT IS USED
Gemini (free)	The newest Gemini Flash model the key can use	First choice on the deployed site
Groq (free)	gpt-oss-120b → Qwen → gpt-oss-20b	Automatically if Gemini errors or times out (15s). A failed provider is skipped for 10 minutes.
Rules engine	Deterministic version of each agent	If no AI answers; cards are labelled "Rules engine"
Replay	The saved AI answers for each demo case	Admin switch (top right) when the network is unreliable

> **GWU campus network:** campus Wi-Fi blocks Google's Gemini address. The deployed site isn't affected (Vercel's servers call Gemini, not your laptop). Running locally on campus, the app falls back to Groq.

8 · UNDER THE HOOD

How it works

Agents and their tools

- **Sales:** 56-day baseline, seasonality for the exact window, category elasticity, real promo-day analogs from M5, channel mix.
- **Finance:** incremental gross profit, basket halo, supplier funding, execution costs, a downside case (−40% uplift), break-even discount solver.
- **Operations:** inventory position, launch requirement vs lead time, DC utilization, labor and online fulfillment, projected fill rate.
- **Legal:** matches 180 active rules (250 including superseded versions) by category, mechanic, channel, state, discount, duration, flags and claim text.

Pipeline

Tools compute, agents reason, guardrails enforce, people sign, the executive decides. Numbers always come from deterministic tools. The model returns a recommendation, confidence, risk, conditions and cited evidence IDs (schema-validated). Guardrails remove made-up citations and enforce hard limits.

Governance (Deliverable 1 §4.2, extended)

- Legal prohibition → Reject; nobody can approve it.
- Missing evidence → invalid output, Review required, never imputed.
- Operations hard stop, or a Finance hold or reject → no proceed.
- Support vs oppose → an explicit conflict, never averaged.
- Confidence = weighted agent confidence × completeness × freshness × (1 − conflict penalty). A human adjustment multiplies that agent's confidence by 0.85.
- Department sign-offs as above; then the executive decides.
- Append-only audit log with a SHA-256 hash chain. Database triggers refuse edits and deletes.

9 · QUESTIONS

Likely questions

QUESTION	ANSWER
Why doesn't every lead review every case?	Review fatigue. Leads sign only where their agent has a concern; clean answers clear automatically. Every lane still shows who owns it, and any lead can step in.
Isn't the human just rubber-stamping the AI?	The record says otherwise. Changing an answer needs a reason and is shown side by side with the AI; the Dashboard tracks how often each department confirmed, changed or blocked its agent.
What if a lead never responds?	After 48 hours a Sales, Finance or Ops lane escalates and the executive may decide without it, with a reason. Legal never escalates.
Can the executive overrule Legal?	No. A Legal rejection or a prohibiting rule locks approval for everyone.
Why is the administrator separate?	Separation of duties: the person who sets the rules and thresholds can't also decide cases under them.
Is this real Walmart data?	Demand patterns are real public Walmart data from Kaggle (M5 daily sales; 45-store weekly sales), calibrated to the FY2026 10-K. Stores, inventory, rules and costs are generated and labelled. No Walmart internal data.
What if the AI hallucinates?	Numbers come from tools; the AI must cite evidence IDs; unknown IDs are removed; hard limits apply after the AI answers (test T12). The department lead is the last check.
Prompt injection?	User text is passed as quoted data, screened for injection patterns, and never treated as instructions (test T13).
Does it meet the 60-second target?	Live runs take about 6–15 seconds; a single-agent re-run about 3–5; reconciliation takes milliseconds (test T14).

10 · HELP

Troubleshooting

- **Every lane says overdue:** the reset was done more than 48 hours ago. Reset again, or raise the sign-off deadline in *Rules & agents*.
- **An agent card says "Rules engine":** no AI provider answered. The result is still valid. Re-run, or switch to Replay.
- **Approve is locked:** by design. The panel says which department is still open; switch to that person.
- **A demo case is already decided:** Jordan → *Rules & agents* → *Demo controls* → *Reset demo cases*.
- **Need a printout:** on any case, *Export packet* → Print / Save as PDF. It lists every department's sign-off.